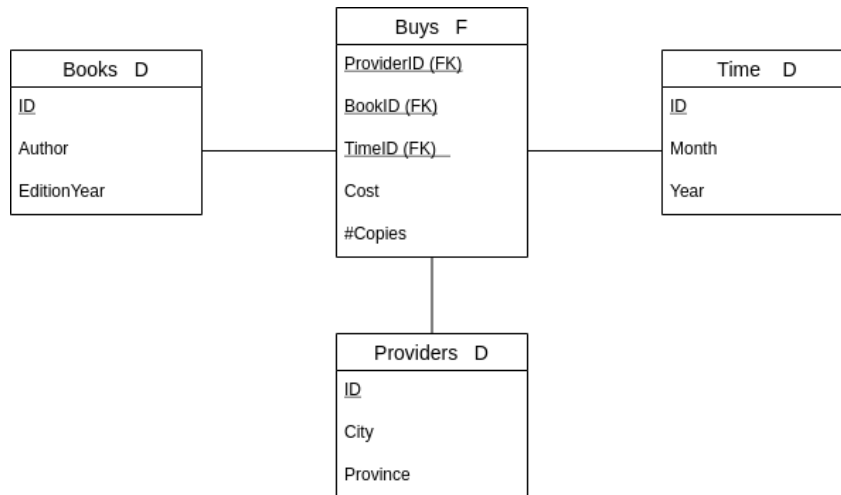


Family name: _____ Given name: _____

20%

1. Given the multidimensional schema, the **sequence of multidimensional operations** below, and the equivalent SQL query, justify if they correspond to each other. If they don't, briefly explain **why and how** the SQL query should be fixed.



- A:=Roll-up(Buys,Providers.city,Sum)
- B:=Roll-up(A, Books.AllBooks,Sum)
- C:=Roll-up(B, Time.AllDays,Sum)
- D:=ChangeBase(C, {Providers})
- E:=Selection(D, city="Barcelona")
- F:=Projection(E, #copies)
- R:=Drill-down(F, Providers.ID)

```

SELECT p.ID, SUM(bu.cost)
FROM Providers p, Buys bu
WHERE p.ID = bu.providerID AND p.city='Barcelona'
GROUP BY p.ID
ORDER BY p.ID;

```

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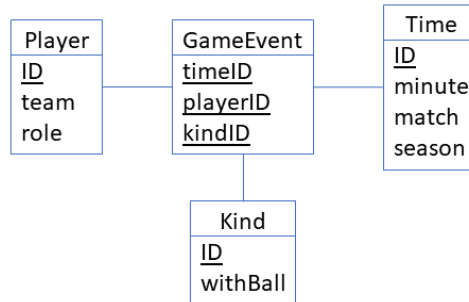
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2. The multidimensional schema below corresponds to events in football matches with and without ball (e.g., pass, tackling) involving players with any role in the team (e.g., goal-keeper, defender). Write a **cube-query (SQL)** over it. If you cannot do it, briefly explain **why**.



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3. Transform the following **run-length encoding** into a matrix of bits where rows correspond to tuples and columns to values. Briefly **justify** whether it is correct or not.

**Run-length
encoded**

x	y	z	t
0	0	1	1
1	0	1	0
1	1	0	1
0	0	1	0
1	1	0	0
1	0	0	0

⇒

Bits matrix

x	y	z	t

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4. Let's suppose we have a table $T(\underline{ID}, A, B, C)$ with 1,000 blocks and 100 records per block. We have a materialized view $V(ID, A, B, C)$ with 100 blocks and also 100 records per block that contains a selection over T by a predicate involving A and B . We receive "UPDATE T SET $C=C+1$ WHERE $Id \% 1000 = 0$,". Assuming that neither the table nor the materialized view have any physical order (i.e., they are not clustered), would it be better a **complete** or an **incremental update** of the view? Briefly justify your answer **numerically**.

Assumptions:

- Statistical independence of attributes
- Uniform distribution of all values in the different data blocks
- No benefit from sequential scan (i.e., no pre-fetching)
- No benefit from consecutive random access of the same block (i.e., no caching)

[illegible]